ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Molecules

Shared by every generator. Pick what your course covers once, and IUPAC naming, Synthesis and future generators all use it, unless you customize a generator separately.

Allowed structures and functional groups

Select all Clear

Generators only build molecules from what's checked.

Hydrocarbons

- [x] Alkane
- [x] Alkene
- [x] Alkyne
- [ ] Aromatic

Functional groups

Core

- [x] Alcohol
- [x] Alkyl halide
- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

Carboxylic acid derivatives

- [ ] Carboxylic acid
- [ ] Acid chloride
- [ ] Anhydride
- [ ] Ester
- [ ] Amide
- [ ] Nitrile

Auxiliary

- [ ] Amine
- [ ] Nitro

Molecule size

Number of carbons in each molecule.

3 to 10

Include rings

Cycloalkanes and other cyclic molecules.

Include heteroatoms in rings

Pyridine, furan, thiophene and similar. Needs Aromatic checked.

Stereochemistry

When included, answers must give correct R/S and E/Z.

Off Include

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base — Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

MOLECULE OPTIONS

Use the shared Molecules settings
Semester 1 · 7 groups · 3–10 carbons · rings on · stereo off · Edit shared

Customize for IUPAC naming only
Changes here won't affect other generators.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents.
3 to 10

Most senior functional group
Chosen from the groups allowed under Molecule options.
Any allowed

Number of principal groups
1 to 2

Number of secondary groups
0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

Next problem →

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents.
4 to 3

Minimum is larger than maximum.

Most senior functional group
Chosen from the groups allowed under Molecule options.
Carboxylic acid

Carboxylic acid isn't in your allowed groups. Allow it

Number of principal groups
0 to 2

Number of secondary groups
0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

Changes save automatically and apply to your next problem

Reset this section to defaults

Next problem →

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

Before you start: choose what to practice

Pick what your course covers. Shared by every generator; change it any time with Settings.

Allowed structures and functional groups

Select all Clear

Generators only build molecules from what's checked.

Hydrocarbons

- [x] Alkane
- [x] Alkene
- [x] Alkyne
- [ ] Aromatic

Functional groups

Core

- [x] Alcohol
- [x] Alkyl halide
- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

Carboxylic acid derivatives

- [ ] Carboxylic acid
- [ ] Acid chloride
- [ ] Anhydride
- [ ] Ester
- [ ] Amide
- [ ] Nitrile

Auxiliary

- [ ] Amine
- [ ] Nitro

Molecule size

Number of carbons in each molecule.

3 to 10

Include rings

Cycloalkanes and other cyclic molecules.

Use defaults

Start practicing →

Next problem →

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base — Coming soon

Molecules

Shared by every generator.

Advanced options

	CARBONS		AMINES (needs Amine checked)	
Secondary (2°) Carbon bonded to 2 carbons · R_2NH	0	to 4	0	to 1
Tertiary (3°) Carbon bonded to 3 carbons · R_3N	0	to 2	0	to 1

Ring topology
Single rings are always included when Include rings is on.
☐ Spiro ☐ Fused ☐ Bridged

Aromatic ring positions
Substituent patterns on disubstituted benzene rings. Needs Aromatic checked.
☐ Ortho (1,2) ☐ Meta (1,3) ☐ Para (1,4)

Functional group count
Total number of functional groups in each molecule.
0 to 3

Functional group distance
How far apart two functional groups are (Warren, *The Disconnection Approach*).
☑ 1,1 ☑ 1,2 ☑ 1,3 ☑ 1,4 ☑ 1,5

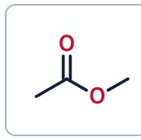
Custom fragments
Every generated molecule must contain all of these fragments.

+ Add fragment



C1CCCCC1

Edit Remove



C(=O)OC

Edit Remove

! Contains an ester, which isn't checked above. Allow ester

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

Practice Problems · Lectures · Account

ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base — Coming soon

Molecules

Shared by every generator.

Add a custom fragment

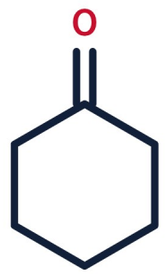
Draw | Enter SMILES

SMILES

O=C1CCCCC1

One fragment per entry, e.g. C1CCCCC1 for a cyclohexane ring.

PREVIEW



Cyclohexanone fragment · Open in Draw to edit

Cancel | Add fragment

eeds Amine checked)

to 1

to 1

Fused · Bridged

Meta (1,3) · Para (1,4)

0 to 3

1,3 · 1,4 · 1,5

+ Add fragment

t checked above.

Allow ester

✓ Changes save automatically and apply to your next problem

Reset this section to defaults

Settings

ALL GENERATORS

Molecules >

GENERATORS

IUPAC naming >

Synthesis Laptop or tablet >

Acid/Base Coming soon

Settings

IUPAC naming

MOLECULE OPTIONS

- (•) **Use shared Molecules**
  Semester 1 · 7 groups · Edit shared
- ( ) **Customize for IUPAC only**

NAMING PROBLEMS

Parent chain length	3	to	10
Most senior group Any allowed			Edit
Principal groups	1	to	2
Secondary groups	0	to	2

PRACTICE

Count skipped problems

✓ Saves automatically

Reset section